

Student Grade Management System

(PROJECT REPORT)

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Design Decisions

The Student Grade Management System is designed as a structured, command-line interface (CLI) application used to track academic profiles, aggregate score evaluation, and generate statistical reports. The system is built on procedural programming and structured encapsulation.

Rather than having the entire code base written in the `main()` function, it is broken down into several functions. This increases transparency and minimizes structural coupling. Data encapsulation is done via the composite data type (`struct student`). This container binds multiple properties such as names, ids, grades and other float data variables into a single logical data entity.

Without risking performance degradation, a fixed capacity was selected based on the problem statement (30 student records). The collections are initialized via a static stack-allocated array, while memory elements transversal is optimized using pointer arithmetic to keep execution deterministic and fast.

Implementation of Core Concepts

1. **Arrays & Structural Aggregation** : Data aggregation is handled by a one-dimensional array of structures initialized directly within the execution scope of `main()` function : `student studentData[30];`. The system uses a variable `studentCount` to maintain state boundaries across execution calls. This avoids reading uninitialized structural garbage memory cells during tabular display phases; If student data entered are only 5 it prevents the system from filling the rest of the empty arrays with 0's or null values.
2. **Function Modularity & Architecture** : The software splits its tasks across eight unique functions distinct from the `main()` function, exceeding the least amount

requested in the problem statement (6). These are the various functions and their roles in the program:

- **inputStudentData ()** : handles console input, clears memory buffers and updates data tables.
- **isStringOnly ()** : Validates input strings to ensure they consist entirely of alphabetic text and spacing characters.
- **isNumericOnly ()** : Validates identification data strings to verify they contain only numeric digits prior to parsing.
- **assignGrades ()** : Implements conditional grading logic and returns corresponding characters.
- **computeGrades ()** : Evaluates summation arithmetic loops and maps structural attributes as well as contains remark assignment with **assignGrades ()** function called in it.
- **displayReports ()** : Render organized data tables using the **<iomanip>** stream library features, It displays the recorded student data enter as well as the grades and their corresponding remarks.
- **displayStatistics ()** : Also uses the **<iomanip>** stream library features, It displays the total number of students, the student with the highest score and their corresponding name, the student with the lowest score and their corresponding name, the average score, the number of students that passed as well as the number of students that failed.
- **searchByID ()** : Implements a sequential linear search algorithm over the initialized structures.

3. **Pointer Address Manipulation & Traversal** : To prove low-level memory handling competency as required, the computation routine uses explicit pointer arithmetic traversal. Instead of using traditional array indexing subscripts **studentData[i]** , the array base address is bound to an indirection pointer variable:

Implemented in **computeGrades ()** function;

```
void computeGrades(student* studentData, int count) {
    if (count == 0) { // To check if no data record is entered.
        std::cout << "No Data Available. Please Input Student Record.\n";
        return;
    }

    // Allocating pointer target addresses for manual traversal offsets

    student* ptr = studentData;

    for (int i = 0; i < count; i++) {
        ptr -> totalScore = ptr -> quizScore + ptr -> midsemScore + ptr ->
examScore;

        //Grade Assignment
        ptr -> grade = assignGrades(ptr->totalScore);
    }
}
```

```

        //Remark Assignment
        if (ptr -> totalScore >= 50) {
            ptr -> remark = "Pass";
        }
        else {
            ptr -> remark = "Fail";
        }
        ptr++; // Physically shifts memory alignment pointer blocks ahead to
consecutive locations.
    }

    std::cout << "Grades have been Calculated" <<std::endl;
}

```

By shifting the target reference via **ptr++** , the program steps forward by the exact size of the student struct in bytes. Structural properties are then accessed and updated using member selection arrow operators (->). This bypasses index lookups and shows a solid understanding of low-level memory management.

Technical Challenges & Fault Tolerance

1. **The Input Stream Interruption Trap** : A major technical challenge appeared when a user entered non-numeric characters into numeric input prompts (like entering letters for a Student ID). In standard C++, this corrupts the **std::cin** stream state bit flags and locks the terminal in an infinite processing loop.
Resolution : The system implements a two-tier validation layer. Inputs are first captured into temporary container strings via **std::getline()** . These strings are validated character-by-character using an iterative routine (**isNumericOnly()**) wrapped around **isdigit()** . Only safe, validated text sequences are converted to integers using **std::stoi()** :

```

while (!isNumericOnly(tempID)) { //while loop with the condition that if input on the
tempID returns false for the isNumericOnly function, user should only enter a valid
ID with numeric characters.
    std::cout << "\nInvalid Input!!!! Try Again: ";
    std::getline(std::cin >> std::ws, tempID);
}
studentData[i].studentID = std::stoi(tempID); //Safe conversion string -> target int
data type

```

2. **The Numerical Identification Leading-Zero Problem** : When student IDs were read directly as integers, any leading zeros (e.g., ID 01234) were automatically dropped by the compiler during parsing. This altered the identification keys.
Resolution : The system uses an advanced validation bridge. IDs are parsed as standard strings during input validation to preserve their exact character structures. Once validated, they are converted to the required integer format. To keep formatting consistent across reports, the printing system uses **<iomanip>** spacing limits, keeping data uniformly aligned without corrupting the underlying database records.

CONCLUSION

The resulting Student Grade Management System completely complies with structural modularity, low-level memory addressing, and exception-trapping specifications

outlined in the project curriculum. By decoupling program data structures from traditional indexing arrays and enforcing rigorous stream filtration traps, the application provides a highly resilient, performant, and secure execution runtime environment inside standard educational compilation environments.